Figure 3.3 illustrates the diverse range of applications of second-generation unmanned aerial vehicle (UAV) systems, highlighting their evolution from military reconnaissance tools to versatile platforms for scientific, environmental, and civilian operations. The figure is divided into four main sectors: military and defense, scientific and environmental, civilian and commercial, and communication and technology, each representing the expanding functionality of drones during this era. In the military domain, UAVs were employed for reconnaissance, surveillance, and electronic warfare, providing real-time intelligence and situational awareness during conflicts such as the Vietnam and Gulf Wars. The scientific and environmental applications segment showcases UAV use in atmospheric research, weather observation, wildlife monitoring, and disaster assessment, offering valuable data for research and humanitarian efforts. The civilian and commercial section highlights drones in aerial photography, mapping, agriculture, and infrastructure inspection, where GPS integration and imaging systems revolutionized industrial efficiency. Finally, the communication and technology segment demonstrates the role of UAVs as platforms for data relay, flight automation testing, and system integration research.